

Mukta Mishra

ASL Gesture Recognition Lab

Screenshots

```
hi: 0.998754  
sup: 0.001246
```

```
hi: 0.990604  
sup: 0.009396
```

```
hi: 0.998230  
sup: 0.001770
```

```
hi: 0.013795  
sup: 0.986205
```

```
hi: 0.229560  
sup: 0.770440
```

Software Issue Fixed

The notebook contained an issue in the model training configuration. The model was originally compiled using Mean Squared Error (MSE) loss and Mean Absolute Error (MAE) metrics, which are typically used for regression problems. However, this lab is a gesture classification task with one-hot encoded labels representing the two gesture classes ("hi" and "sup").

To correct this issue, I changed the model compilation step to use categorical crossentropy loss and accuracy as the evaluation metric. This makes the training process appropriate for a classification problem and provides a more meaningful measure of model performance.

Changes Made for Consistency

After changing the loss function and metric, I updated the rest of the notebook to remain consistent with the classification-based approach. Specifically:

- Changed the training metric from MAE to accuracy.
- Updated the metric plotting section to use accuracy and validation accuracy.
- Regenerated the TensorFlow Lite model and exported a new model.h file for deployment.